

1 hour on lecture notes (proving hall)

Problem 1 (Z5581CD0)

For every subset of 1007 elements there are 1008 ways to select one of the remaining numbers to be added to it to create a subset of 1008 elements that is a superset of the original.

For every subset of 1008 numbers we have 1008 ways to remove one element and get a subset of the original set with 1007 elements.

Therefore, regular hall.

Problem 2 (ZB9281DC)

We show that Hall's Condition is met.

FTSOC, let this not be the case meaning there is a subset of k elements from A that are matched to less than k elements in B .

If $k = 0$, the size of no subset is less than zero so impossible.

If $1 \leq k \leq \frac{n}{2}$, then any element in A is matched to at least $\frac{n}{2}$ elements in B .

If $\frac{n}{2} < k \leq n$, we let $m < k \leq n$ be the subset of elements in B that are at least matched to one of the k elements in A . This means that there is at least one remaining element in B that is not paired to any of these k elements in A , instead being paired to some of the $n - k < \frac{n}{2}$ remaining elements in A .

However, this is impossible due to minimum degree $\frac{n}{2}$.

Problem 4 (Z05797B3)

We duplicate the left-hand-side. For every element x , we create an identical element x' adjacent to the same set of elements in B . Hall's condition is met with every subset S of the LHS adjacent to at least $|S|$ members of the RHS. Therefore, by Hall, there is a 1-to-1 matching of elements here.

Note that x' and x still may not be mapped to the same element. Therefore, solving the original problem, we match x with what it was mapped to as well as what x' was mapped to to be done.

6 clubs; 2 hours

Problem 5 (ZB1FC962)

Let $A = \{a_1, a_2, \dots\}$; $B = \{b_1, b_2, \dots\}$. We can have a_1 adjacent to all the elements in B . However, $\forall i > 1$, we have a_i adjacent to b_{i-1} . Clearly, every finite subset S of A that includes a_1 is adjacent to infinitely many elements in B and all finite subsets S that exclude a_1 is adjacent to exactly $|S|$ neighbors in B .

However, there is no matching of A to B . For any element b_i which a_1 is matched to, a_{i+1} cannot be matched to.

Problem 6 (ZF13143F)

We claim the answer is all $k \leq 500$.

We first prove the bound. If $k > 500$, then we remove the $k \cdot k$ square in the top right. Since there must be exactly one rook per row, in each of the first k rows, exactly one rook occupies a distinct column. By Pigeonhole, there are only $n - k < k$ available columns so impossible.

We now prove $k \leq 500$ is attainable. Consider a bipartite graph between the 1000 rows $r_1, \dots, r_{1000}$ and the 1000 columns $c_1, \dots, c_{1000}$. We have an edge between r_i, c_j if the cell (i, j) is not deleted meaning we can place a rook there.

There must be exactly one rook in each row and one in each column, so this is possible if and only if there is a bijection from rows to columns using only the edges (saying this always feels awkward associating edges with a function, is this the best wording?).

When $k \leq 500$, all vertices still have degree at least 500 since at most 500 squares were deleted from each row and column. Therefore, by Problem 2, we are done. (Can I quote P2 directly in a contest?)

11 clubs; 2.5 hours

Problem 7 (12PTNMB3)

We consider a bipartite graph with $4n$ vertices, $2n$ on each side. The left side has $2n - 1$ vertices representing each of the $2n - 1$ days. The right side has $2n$ vertices representing each team. We draw an edge between a day-team pair if that team wins on that day.

The problem asks us to find an injective function from each day to a different team. This is possible if and only if Hall's criterion is met. FTSOC, let Hall's criterion not be met, meaning during a subset of k days, there are less than k teams have won at least one match and $2n - k$ teams that have won zero matches.

For this selection of k days, we have most $k - 1$ "winning" teams that win at least one game meaning at least $2n - k + 1$ "losing" teams who never win a game.

All the $\binom{2n - k + 1}{2}$ pairings of teams who never win a game must happen on the $2n - 1 - k$ days that are not selected for if two "losing" teams get matched together there is a win. During the unselected days, however, each "losing" team may play at most 1 other "losing" team per day so it'll take $2n - k$ unselected days for this to happen. (!)

Since Hall's criterion is met, we are done.

Problem 8 (Z685102E)

FTSOC, let there be a valid configuration where no subset of a rows and b columns for $a + b \geq n + 1$ has their intersection contain only blue squares.

We consider a bipartite graph between each of the n rows to each of the n columns. We connect an edge between a row to a column if there is a blue square there.

Sketch: Hall's criterion is not met so must be a rows and b columns for $a + b \geq n + 1$ where their intersection contains only blue (!).

21 clubs; 3.5 hours

Problem 9 (WOOTPO46)

We claim that $k = 3$.

We start by proving that $k < 3$ is not enough. Label each cell in the initial grid by an ordered pair containing their (0-indexed) row number followed by their column number.

(0, 0)	(0, 1)	...	(0, m-1)
(1, 0)	(1, 1)	...	(1, m-1)
...	...	...	...
(n-1, 0)	(n-1, 1)	...	(n-1, m-1)

Note that the labeling remains fixed to identify the initial position of a cell even as we perform moves.

We show that it is impossible to reach this configuration where the corners are cyclically rotated in at most two moves.

(n-1, 0)	(0, 1)	...	(0, 0)
(1, 0)	(1, 1)	...	(1, m-1)
...	...	...	...
(n-1, m-1)	(n-1, 1)	...	(0, m-1)

We both have cells that need to change rows and cells that need to change columns, so we permute exactly once on the rows and once on the columns. WLOG, let us permute the rows first. After the first move, each cell is locked in its column for the second move meaning each cell needs move to the column it needs to be in at the end. However, this is not possible as $(0, 1) \dots (0, m - 1)$ must not move but $(0, 0)$ needs to move to the end.

We may go in reverse. We show that 3 moves are enough to reach the start from any permutation.

Claim: In one move permuting on rows, we can get to a board where, for every column, there is exactly one of each number $0, \dots, n - 1$ in the first coordinate of the labels.

We prove this by induction on how many columns there are. Note that we can permute the rows (and only rows) multiple times in a row, taking intermediate steps and only lock in the final state.

Base Case: 1 column. The starting position works.

Inductive Step $n = k - 1 \rightarrow n = k$: We prove that we can construct at least one column where there is exactly one of each number $0, \dots, n - 1$ in the first coordinate. This is simply regular hall. After that, we use the induction hypothesis to process the remaining $k - 1$ columns.

Conclusion: By mathematical induction the claim is proven.

We perform the move that achieves this for our first move. For our second move, we permute the columns such that all the cells go to the right row, which is possible since there is one cell from each row in each column. For our last move, since every cell is in the right row, we permute the rows such that the cells go to the correct column.

Problem 11 (10SLC2)

We claim that the answer is $2^{n-2} + 1$.

To prove this is the lower bound, notice the set of all binary strings of length n that start with 0 and end with 1 may never satisfy the problem statement as nothing in the first column can be 1 and nothing in the last can be 0.

We now prove that any set of $2^{n-2} + 1$ numbers have a diverse subset.

Case 1: There are no columns filled with zeroes.

Consider a bipartite graph from rows to our $2^{n-2} + 1$ flags. We connect an edge between row i to every flag where the i -th bit of the flag is a one. If we can find an injective function from the edge to the flags on the edges, then we'd have a subset of flags with the main diagonal all one.

We show that Hall's Criterion is met. FTSOC, let it be not, meaning for k rows, there are less than k flags adjacent to at least one row.

If $k = 1$, there are no columns filled with zeroes so impossible.

If $k \geq 2$, if this was met, then at least $2^{n-2} + 1 - (k - 1) = 2^{n-2} + 2 - k$ of our flags have no ones in the k rows. However, there are only 2^{n-k} such unique flags. $2^{n-k} < 2^{n-2} + 2 - k$ for $k > 2$.

However, for $k = 2$, the equality is met meaning all such unique flags appear by Pigeonhole. WLOG, let the two rows be the first and the second rows with no ones. However, in this case, we may get a board with diagonal filled with zeroes by construction or hall.

Case 2: There exists a column filled with zeroes.

There must not exist a column filled with ones for otherwise we may only have 2^{n-2} distinct binary strings satisfying this. Therefore, we apply hall the same way as above except switching the zeros and ones around.

31 clubs; 6 hours